

Seat No:

SET	P
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PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR UNIVERSITY
B.Tech End Semester Examination (CBCS)

Subject: Computer Networks & Security · Pattern: CBCS Pattern

Day & Date: Thursday, September 17, 2026 **Duration:** 180 Minutes **Max. Marks:** 70 Marks

Instructions for Candidates:

1. Q.1 is compulsory. Mention the question paper set (**P**) clearly on top of the answer booklet.
2. Figures to the right indicate full marks assigned to each question.
3. Assume suitable data wherever necessary and state your assumptions explicitly.
4. Use of programmable calculators or unauthorized electronic communication devices is strictly prohibited.
5. Standard University Marking Scheme

Q.1 Choose the correct alternatives for the following questions. [2 Marks]

1. Which layer of the OSI model is responsible for routing? What is $O(\log n)$ routing cost?
 - a) Network Layer
 - b) Transport Layer
 - c) Data Link Layer
 - d) Physical Layer
2. What is the formula for Shannon Channel Capacity? Consider $C = B \log_2(1 + \text{SNR})$.
 - a) $B \log(1+S/N)$
 - b) $2B \log(M)$
 - c) None of these
 - d) All of the above

SECTION – I (Theory & Core Concepts) [28 Marks]

Q.2 Answer the following questions (Attempt Any Four): [16 Marks]

- a) Explain **Distance Vector Routing** and Count to Infinity problem with state diagram. [4]

Q.3 Answer the following questions in detail (Attempt Any Two): [12 Marks]

- a) Explain **Distance Vector Routing** and Count to Infinity problem with state diagram. [4]

— END OF QUESTION PAPER (P) —